Selection and Application Instructions for AILEIKE ALKW851B Dry Body Temperature Verification Furnace

The AILEIKE ALKW851B dry body temperature verification furnace is a high-performance, portable temperature verification device. It features a liquid crystal touch display screen, supports switching between Chinese and English interfaces, and offers simple and intuitive operation. With stable overall performance, convenient operation, and no medium pollution, it combines laboratory verification accuracy with on-site calibration portability, making it highly practical.

Common aliases include: dry body furnace, dry well furnace, temperature verification furnace, dry temperature verification furnace, portable temperature verification furnace, etc.

1. Technical indicators

name	Dry body temperature verification furnace	trap depth	150mm
Model number	ALKW851B	trap opening size	Φ32mm
temperature range	35~650℃	Measure the diameter of the hole	Φ6mm Φ8mm Φ10mm Φ12mm
Heat source	electric heating	Number of measuring holes	4 holes
Control method	LCD touch screen	Power	600W
Instrument accuracy	Level 0.1	Power supply	220V 50Hz
display resolution	0.01℃	Overall dimensions	360×165×330mm
stability	±0.1℃	weight	8.5kg
uniformity	±0.1℃		

II. Applicable Scenarios

1. Industrial site inspection: factories, power plants, chemical industries, metallurgy, shipbuilding and marine applications, and other such places.

2. Calibration for various temperature sensors: Calibrate PT100 thermal resistors, various types of thermocouples, temperature transmitters, and small temperature meters.

3. Batch calibration: Multiple probes can be calibrated simultaneously through four temperature measurement holes, enhancing calibration efficiency.

4. Periodic equipment calibration: Regular annual inspections are conducted on temperature measurement instruments associated with mechanical equipment, pipelines, boilers, and ovens.

5. Third-party metrology fieldwork: Conduct temperature testing on-site, retain calibration data, and fill out calibration records.

III. Calibratable products

1. Thermocouple

It can calibrate thermocouples of type K, J, E, T, etc., with a measurement range not exceeding 650℃.

2. Thermal resistance

Compatible with PT100, PT1000, and Cu50, suitable for temperature measurement within the range of 35~650℃.

3. Temperature transmitter and display device

4~20mA, 0~10V output temperature transmitter, as well as the supporting digital temperature display and recorder.

4. Plug-in temperature instrument

Bimetallic thermometer, portable temperature probe, surface temperature probe.



IV. Non-verifiable Products and Instructions

1. S-type, R-type, and other high-temperature thermocouples are not applicable.

2. Sensors with temperatures below 30℃ or above 650℃ cannot be calibrated.

3. Non-intrusive devices: Thermal imaging cameras and radiation thermometers. These devices require calibration using a blackbody furnace.

4. Sensors with a protective tube diameter greater than 12mm cannot be placed in the measurement and

control system for calibration, and require special customization of the furnace core aperture.

5. Insufficient probe length (such as a short-stem thermistor less than 70mm) results in significant heat dissipation effects and excessive temperature measurement errors.

V. ALKW851B temperature rise and fall time

warm-up time		cooling time	
temperature range	time-consuming	temperature range	time-consuming
Normal temperature 25~100℃	11 minutes	650～300℃	42 minutes
100～300℃	8 minutes	300～100℃	43 minutes
300～650℃	16 minutes		

VI. Conclusion

The AILEIKE ALKW851B Dry Body Temperature Calibration Furnace is a high-precision, portable temperature calibration device. It boasts a wide temperature measurement range of 35~650℃, stable temperature control, and uniform temperature distribution. It can be adapted to the calibration of various thermocouples, thermal resistors, and temperature measurement instruments. The device exhibits excellent temperature rise and fall efficiency, combining the advantages of low power consumption and lightweight design. It is suitable for various scenarios such as industrial inspections, equipment annual inspections, and third-party metrology, effectively enhancing the efficiency of calibration work. When using, it is necessary to strictly adhere to specifications, avoid overtemperature use, ensure the probe insertion length, and block idle holes to ensure accurate and reliable calibration data. It is an indispensable verification device for industrial temperature measurement calibration.